UNIVERSAL WEIGHTED-LAPLACIAN CONES AND THE MAXIMAL-CUSP SPECTRAL LINK

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Abstract. For each order we construct a scale-invariant cone of simple lacunary spectra realized, at regular points with the strong spectral property, by every connected weighted graph. A toric blow-up of the rooted-forest coefficient map gives the boundary scale–shape determinant $n/(n-1)$. We also obtain the sharp weighted-path eigenratio and an $O((\log n)/n)$ upper bound for its normalized spectral volume.

The geometric part identifies the projectivized tangent link of the first $2g-3$ positive eigenvalues at the spectrally maximal cusp of genus-g moduli space. The link is a finite union of normalized weighted-Laplacian spectral images of complexity-one stable types; zero conductances encode arbitrary relative pinching scales. We make the required surface-uniform Burger comparison explicit. Outside a compact semialgebraic null skeleton, every feasible simple direction lifts exactly at every sufficiently small scale; weak-Arnold-hypothesis collision directions lift with multiplicity preserved.

The link has sharp squared distance $(g-3)/(3(g-1)(2g-3))$ from the equal-spectrum direction. Hence a sharp tangent cone is absent for $g \geq 4$, whereas in genus 3 every positive triple ray, including repetitions, has a fixed-curvature exact lift. A finite-size bipartite Ramanujan construction supplies, in every genus, a full-dimensional exact wedge of condition number below $18 + 12\sqrt{2}$; every target in it has continuum many nonisometric realizations with the same entire sub-1/4 spectrum. We further give an existential-real membership formulation, graph-by-graph nonmembership certificates, and an exact-arithmetic Frucht-fold certificate. All surface conclusions beyond the tangent link require a regular or essential graph preimage. The finite-scale inverse problem, including the residual collision skeleton and behavior near 1/4, remains open.

1. Introduction

The inverse eigenvalue problem for a weighted graph Laplacian asks which lists

$$0 = \lambda_1 < \lambda_2 \leq \cdots \leq \lambda_n$$

can occur when the positive edge weights of a fixed connected graph are allowed to vary. Even for paths, the row-sum-zero constraint makes this problem markedly different from the unconstrained symmetric inverse eigenvalue problem. Recent work of Fallat–Gupta–Lin [15] and Catral–Fallat–Gupta–Lin [8] developed Jacobian and strong-property tools for this setting, while Erchenko–Jakobson–Tsypin [14] connected the realizable regions to small eigenvalues of degenerating hyperbolic surfaces. Mukherjee [21] proved approximate finite spectral prescription for constant-negative-curvature manifolds when topology and the curvature scale may vary; in dimension two that argument also uses the classical collar graph limit. Our surface results instead keep both the genus and curvature -1 fixed and give exact realization on specified near-cusp regions.

The usual direction is graph by graph: one searches for inequalities or constructs a weighting adapted to the particular support. Our first result instead isolates a spectral region common to

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every connected support. For $0 < \delta < 1$ set

$$\mathcal{L}_n(\delta) = \left\{ 0 < \lambda_2 < \cdots < \lambda_n : \frac{\lambda_i}{\lambda_{i+1}} < \delta \quad (2 \le i \le n-1) \right\}. \tag{1}$$

Theorem 1.1 (Universal lacunary cone)**.** *For every $n \ge 2$ there is $\delta_n \in (0,1)$ such that*

$$\mathcal{L}_n(\delta_n) \subseteq \mathcal{E}_{>0}(G)$$

for every connected graph G on n vertices. Every list in this cone has a strict realization which is a regular point of the spectral map and has the SSPWL*.*

As of 4 September 2026, we are not aware of an earlier support-independent open cone of this fixed spectral form lying in the strict realization set of every connected graph on n vertices. Existing strong-property results supply graph-dependent local spectral neighborhoods; the common cone is the additional assertion here.

The mechanism behind Theorem 1.1 is the main point. Choose a spanning tree and order its edges so that the first k edges form a connected tree on $k+1$ vertices. If successive edge weights are separated by independent small ratios, the rooted-forest formula has a unique leading monomial in every coefficient. Dividing out those monomials produces an analytic map at a toric corner. Its shape Jacobian is diagonal there, and the determinant of the full scale–shape Jacobian telescopes to

$$2\prod_{k=2}^{n-1}\left(1 - \frac{1}{k^2}\right) = \frac{n}{n-1}.$$

This supplies an exact local chart, including anisotropic lacunarity. A second implicit-function argument turns the zero weights on the remaining edges of the graph into small positive weights without changing the characteristic polynomial.

Our second theme is quantitative thinness. For the path P_n, a classical optimal-scaling argument of Forsythe–Straus [16] specialized to the type-A_{n-1} Cartan matrix gives

$$\frac{\lambda_n}{\lambda_2} \ge \cot^2\!\left(\frac{\pi}{2n}\right), \tag{2}$$

with equality precisely for equal edge weights. The inequality itself is classical in this matrix-scaling guise; our application is to the spectral volume $P(G)$ introduced in [14]. An exact beta integral for the simplex region cut out by (2) yields an explicit upper bound $P(P_n) \le U_n$, with

$$U_n = \frac{\pi^2}{4n}(\log n + \gamma_{\mathrm{E}}) + O\!\left(\frac{(\log n)^2}{n^2}\right).$$

Thus a universal nonempty cone can coexist with vanishing normalized volume.

We now turn to the geometric problem that motivated [14]: prescribing the first $2g-3$ eigenvalues of a closed curvature--1 surface of genus g. Put

$$N = 2g-2, \qquad m = N-1 = 2g-3,$$

and, for a closed surface X, enumerate

$$0 = \lambda_0(X) < \lambda_1(X) \le \lambda_2(X) \le \cdots$$

and write

$$\Lambda_g(X) = (\lambda_1(X), \ldots, \lambda_m(X)), \qquad p(X) = \frac{\Lambda_g(X)}{\sum_{j=1}^{m}\lambda_j(X)}.$$

The number m is the full range below $1/4$ allowed by Otal–Rosas [22]. We call

$$\mathcal{T}_g = \left\{ \lim_{k\to\infty} p(X_k) : \lambda_m(X_k) \longrightarrow 0 \right\}$$

the *spectrally maximal-cusp tangent link*. This terminology refers to the maximal collapsing cluster; it does not require every Deligne–Mumford node to open at one scale.

Let $\mathfrak{G}_g$ be the finite collection of complexity-one stable genus-g pseudographs described in Section 9. For $\Gamma \in \mathfrak{G}_g$, combine parallel nonloop cuff conductances, ignore loops in the graph energy, and allow the effective conductances w to vanish. If $0 = \mu_1 \le \cdots \le \mu_N$ is the spectrum of $L_\Gamma(w)$, put

$$\psi_\Gamma(w) = \frac{(\mu_2, \dots, \mu_N)}{\operatorname{tr} L_\Gamma(w)}.$$

Theorem 1.2 (Exact maximal-cusp tangent link)**.** *For every $g \ge 2$,*

$$\mathcal{T}_g = \bigcup_{\Gamma \in \mathfrak{G}_g} \{\psi_\Gamma(w) : w \ge 0,\ \operatorname{tr} L_\Gamma(w) > 0\}.$$

The link is compact and semialgebraic. Zero conductances are essential: they record multiscale pinching and may create additional zero coordinates in the limiting direction.

On the simple positive part of the link, let $\mathcal{U}_g$ consist of the directions having at least one normalized-regular graph preimage, and set

$$\mathfrak{B}_g = \mathcal{T}_g \setminus \mathcal{U}_g.$$

Thus a simple point lies in $\mathfrak{B}_g$ precisely when every graph preimage is critical; repeated and coordinate-wall directions are also placed in the skeleton without assigning a derivative to the ordered eigenvalue map.

Theorem 1.3 (Generic exactness)**.** *The set $\mathcal{U}_g$ is relatively open and semialgebraic in the strict ordered simplex (rather than merely relatively open in $\mathcal{T}_g$), while $\mathfrak{B}_g$ is compact and semialgebraic with*

$$\dim \mathfrak{B}_g \le m - 2, \qquad \mathcal{H}^{m-1}(\mathfrak{B}_g) = 0.$$

Here $\mathfrak{B}_2 = \varnothing$ and $\dim \varnothing = -\infty$. For every $p \in \mathcal{U}_g$ there is $s_0(p) > 0$ such that every sp, $0 < s < s_0(p)$, is exactly the first m positive-eigenvalue list of a closed genus-g hyperbolic surface. If $p \notin \mathcal{T}_g$, then no sp is realizable for all sufficiently small $s > 0$.

The regularity hypothesis is sufficient, not necessary. A form-valued version of the Colbois–Colin de Verdière essential-family argument [10] also lifts collision directions admitting the weak Arnold hypothesis (WAH); the strong Arnold hypothesis (SAH), and in particular the SSPWL, implies WAH. Directions not settled by either mechanism remain confined to the lower-dimensional skeleton.

Theorem 1.4 (Conditioned wedge and maximal-prefix nonrigidity)**.** *For every $g \ge 2$ there are a nonempty open set $V_g \subset \{0 < \eta_1 < \cdots < \eta_{2g-3}\}$ and $s_g > 0$ such that, for every $\eta \in V_g$ and $0 < s < s_g$,*

$$\frac{\eta_{2g-3}}{\eta_1} < 18 + 12\sqrt{2},$$

and continuum many pairwise nonisometric closed genus-g surfaces have $s\eta$ as their entire sub-$1/4$ spectrum. For each fixed s, almost every target in sV_g has a regular marked realization with local isolevel dimension $4g - 3$.

The next eigenvalue is always larger than $1/4$, so the outside gap can be made arbitrarily large while retaining the displayed uniform internal bound. In addition, for large g one common exact maximal prefix appears in at least $\exp\{g \log g - O(g)\}$ canonically distinguishable all-pants thin regions. The word *Ramanujan* refers here only to the finite cubic bipartite dual multigraph supplied by [20]; no automorphic or Selberg-type assertion is involved.

The topology of a stable genus-g graph forces a sharp quantitative gap. Write

$$b_m = (1/m, \dots, 1/m).$$

Theorem 1.5 (Sharp cusp gap and the genus-3/4 transition)**.** *For every $g \geq 3$,*

$$\lim_{\varepsilon \downarrow 0} \inf_{\lambda_{2g-3}(X) \leq \varepsilon} \|p(X) - b_m\|_2^2 = \frac{g-3}{3(g-1)(2g-3)}.$$

Consequently, for every $g \geq 4$ and every $r < \sqrt{(g-3)/(3(g-1)(2g-3))}$, the radius-$r$ angular cone about the equal-spectrum ray is absent sufficiently near the origin; the displayed radius is sharp at the tangent level. In contrast, for $g = 3$ and every $0 < a \leq b \leq c$, there is $s_0(a,b,c) > 0$ such that every $s(a,b,c)$, $0 < s < s_0(a,b,c)$, is exactly the first three positive eigenvalues of a closed genus-3 hyperbolic surface, with multiplicities preserved.

If

$$\kappa_g^* = \inf_{p \in \mathcal{U}_g} \frac{p_m}{p_1}, \qquad N = 2g-2,$$

then the sharp variance theorem and the conditioned wedge give, for $g \geq 3$,

$$\frac{5N - 8 + 4\sqrt{(N-1)(N-4)}}{3N} \leq \kappa_g^* < 18 + 12\sqrt{2}.$$

The lower bound tends to 3; this bracket does not determine κ_g^*.

Colin de Verdière's property A_4 gives the corresponding unnormalized constant-curvature finite-spectrum prescription in genus three, with four eigenvalues including the zero mode and with repetitions allowed [11, Theorem 1.3 and §2a]. That theorem does not fix the curvature normalization. The statement here keeps curvature equal to -1 and realizes every sufficiently small point on each ray by a direct K_4 essential-family argument; its role is to identify the zero-gap endpoint of the maximal-cusp link theorem.

The universal cone remains a concrete regular chart inside this larger picture.

Theorem 1.6 (Exact lacunary prescription on a fixed-genus surface)**.** *For every $g \geq 2$ there is $\varepsilon_g > 0$ with the following property. If*

$$0 < \rho_1 < \cdots < \rho_{2g-3}, \qquad \frac{\rho_j}{\rho_{j+1}} < \varepsilon_g \quad (1 \leq j \leq 2g-4),$$

then there is $s_0 = s_0(\rho) > 0$ such that, for every $0 < s < s_0$, a closed hyperbolic surface X_s of genus g has

$$\lambda_j(X_s) = s\rho_j, \qquad 1 \leq j \leq 2g-3, \qquad \lambda_{2g-2}(X_s) > \frac{1}{4}.$$

The final layer is effective in the sense of real algebraic geometry. For a rational direction p, tangent-link membership has a polynomial-size existential-real encoding. For fixed genus, standard fixed-dimensional quantifier-elimination bounds give polynomial bit complexity. Off the link, strict positivity of a trace-moment error polynomial on each compact weight simplex gives a finite Putinar–Lasserre sums-of-squares certificate. These are upper bounds and certificates only: no hardness result and no uniform SOS degree bound is asserted.

The logical scope is worth recording explicitly. The tangent-link theorem is a complete asymptotic classification of projectivized maximal collapsing clusters. Exact statements at positive scale require either a regular graph preimage or an essential form-valued family. We do not solve the full finite-scale problem of [14, Question 1], classify the residual collision skeleton, or control spectral continuation through the threshold 1/4.

The paper is organized as follows. Sections 2– 4 prove the universal graph theorem and record a sharp generic algebraic-degree bound for trees. Sections 5– 6 give the independent path obstruction and volume consequence. Sections 7–8 establish the exact graph-to-surface transfer.

Sections 9– 14 prove the tangent-link, generic-exactness, conditioned maximal-cluster, finite-prefix nonrigidity, essential-collision, sharp-gap, and genus-three theorems. Section 15 develops the algebraic decision and SOS certificates, and Section 16 isolates the unresolved singular directions and the global continuation problem.

2. Weighted Laplacians and spectral coordinates

Let $G = (V, E)$ be a simple graph on $V = [n]$. Give each edge $e = uv$ an orientation and put $b_e = \mathbf{e}_u - \mathbf{e}_v$. For $w \in \mathbb{R}^E_{>0}$ its strictly weighted Laplacian is

$$L_G(w) = \sum_{e \in E} w_e b_e b_e^{\mathsf{T}}. \tag{3}$$

The orientation is immaterial. We denote by $\mathcal{E}_{>0}(G)$ the set of ordered positive spectra realized by such weights. When G is connected, $L_G(w)$ is positive semidefinite with kernel $\mathbb{R}\mathbf{1}$.

Put $m = n - 1$ and write the characteristic polynomial in the form

$$\det(tI_n + L_G(w)) = t \prod_{i=2}^{n} (t + \lambda_i) = \sum_{k=0}^{m} q_k(w) t^{n-k}, \qquad q_0 = 1. \tag{4}$$

Thus $q_k = e_k(\lambda_2, \dots, \lambda_n)$. A realization is *coefficient-regular* if the map

$$w \longmapsto (q_1(w), \dots, q_m(w))$$

has rank m. For a simple spectrum this is equivalent to regularity of the ordered spectral map.

We will repeatedly use the following elementary coordinate system on the positive coefficient orthant. For $2 \le k \le m$ define

$$\vartheta_k(q) = \frac{q_k q_{k-2}}{q_{k-1}^2}. \tag{5}$$

Then

$$q_k = \vartheta_k \frac{q_{k-1}^2}{q_{k-2}}, \tag{6}$$

so $q \mapsto (q_1, \vartheta_2, \dots, \vartheta_m)$ is a diffeomorphism of the positive orthant.

Let $u_2, \dots, u_n$ be orthonormal eigenvectors for a simple nonzero spectrum. The eigenvalue derivative is

$$\frac{\partial \lambda_i}{\partial w_e} = (b_e^{\mathsf{T}} u_i)^2. \tag{7}$$

The matrix on the right, with rows indexed by $i = 2, \dots, n$, is the spectral Jacobian with respect to the edge weights. Since the Jacobian from the simple eigenvalues to their elementary symmetric functions is nonsingular, coefficient regularity is equivalent to full row rank of this spectral Jacobian. It is not the verification matrix of [8]; rather, the implication from this regularity to the SSPWLfollows from [8, Theorem 3.8].

3. Spanning forests and toric corner charts

The coefficient blow-up rests on the rooted-forest formula. For a forest F, let $\pi(F)$ denote all components of the spanning forest (V, F), including isolated vertices.

Lemma 3.1 (Rooted-forest coefficients)**.** *For $0 \le k \le n - 1$,*

$$q_k(w) = \sum_{\substack{F \subseteq E(G),\ |F| = k \\ F \text{ a forest}}} \left(\prod_{C \in \pi(F)} |C| \right) \prod_{e \in F} w_e. \tag{8}$$

Proof. Let B be an oriented incidence matrix and $W = \operatorname{diag}(w_e)$, so $L = BWB^{\mathsf{T}}$. The coefficient q_k is the sum of the principal k-minors of L. For $S \subseteq V$, $|S| = k$, Cauchy–Binet gives

$$\det L[S,S] = \sum_{\substack{F \subseteq E(G) \\ |F|=k}} \det B[S,F]^2 \prod_{e \in F} w_e.$$

The determinant is nonzero precisely when F is a forest and every component contains exactly one vertex of $V \setminus S$; then its square is one. For fixed F, choosing $V \setminus S$ is therefore the same as choosing one root in each component, in $\prod_{C \in \pi(F)} |C|$ ways. Summing over S proves (8). This is the rooted-forest specialization of the all-minors matrix–tree theorem; see [9]. □

Let T be a tree on n vertices. Fix an ordering

$$E(T) = \{e_1, \dots, e_m\} \tag{9}$$

such that the first k edges form a connected tree on $k+1$ vertices for each k. Indeed, root T, list the nonroot vertices in a parent-before-child order, and let e_k join the kth listed vertex to its parent. Equivalently, the reverse order is a pendant-edge deletion order.

Introduce the toric weight coordinates

$$w_1 = s, \qquad w_j = s x_1 \cdots x_{j-1} \quad (2 \le j \le m), \tag{10}$$

where $s > 0$. For $I \subseteq [m]$ and $1 \le r \le m-1$, put

$$\nu_r(I) = |\{i \in I : i > r\}|.$$

If $|I| = k$, then

$$\nu_r(I) \ge (k-r)_+ = \nu_r([k]). \tag{11}$$

Equality for every r holds only for $I = [k]$. Consequently, after (10), every monomial in q_k is divisible by

$$s^k X_k(x), \qquad X_k(x) = \prod_{r=1}^{k-1} x_r^{k-r}.$$

The prefix forest $\{e_1, \dots, e_k\}$ has one component of size $k+1$ and all other vertices isolated. Lemma 3.1 and (11) therefore give the exact factorization

$$q_k^T(s,x) = (k+1) s^k X_k(x) A_{T,k}(x), \qquad A_{T,k}(0) = 1, \tag{12}$$

where $A_{T,k}$ is a polynomial depending on this fixed prefix ordering. We set $X_0 = A_{T,0} = 1$.

Combining (12) with (5) yields

$$\vartheta_k^T(x) = \gamma_k x_{k-1} \frac{A_{T,k}(x) A_{T,k-2}(x)}{A_{T,k-1}(x)^2}, \qquad \gamma_k = \frac{(k-1)(k+1)}{k^2} = 1 - \frac{1}{k^2}. \tag{13}$$

Thus the blown-up shape map

$$\Phi_T(x) = (\vartheta_2^T(x), \dots, \vartheta_m^T(x))$$

extends analytically across the corner and

$$D\Phi_T(0) = \operatorname{diag}(\gamma_2, \dots, \gamma_m). \tag{14}$$

For the analytically extended blown-up map, including the scale coordinate gives the exact boundary determinant

$$\det D_{(s,x)}(q_1, \vartheta_2, \dots, \vartheta_m)\big|_{(s,x)=(s,0)} = 2 \prod_{k=2}^{m} \left(1 - \frac{1}{k^2}\right) = \frac{m+1}{m} = \frac{n}{n-1}. \tag{15}$$

Here the derivative in s is 2 at the corner; the determinant is independent of the positive boundary scale. Formula (15) is the transversality mechanism of the paper.

4. A universal lacunary realization cone

Write a target list in decreasing order as

$$\ell_1 = \lambda_n, \quad \ell_2 = \lambda_{n-1}, \quad \dots, \quad \ell_m = \lambda_2$$

and set

$$\ell_1 = \tau, \qquad \ell_j = \tau z_1 \cdots z_{j-1} \quad (2 \le j \le m). \tag{16}$$

Then $z_j = \ell_{j+1}/\ell_j$, so $0 < z_j < \delta$ is exactly the lacunarity condition. If $p_k = e_k(\ell_1, \dots, \ell_m)$, the same monomial argument as above, now with every subset coefficient equal to one, gives

$$p_k(\tau, z) = \tau^k X_k(z) B_k(z), \qquad B_k(0) = 1. \tag{17}$$

We set $B_0 = 1$. Hence

$$\vartheta_k^{\mathrm{sp}}(z) = z_{k-1} \frac{B_k(z) B_{k-2}(z)}{B_{k-1}(z)^2}. \tag{18}$$

The map

$$\Phi_{\mathrm{sp}}(z) = (\vartheta_2^{\mathrm{sp}}(z), \dots, \vartheta_m^{\mathrm{sp}}(z))$$

is analytic near zero, with $\Phi_{\mathrm{sp}}(0) = 0$ and $D\Phi_{\mathrm{sp}}(0) = I_{m-1}$.

Proposition 4.1 (Tree lacunary chart)**.** *For every tree T on n vertices there is $\delta_T > 0$ such that every list in $\mathcal{L}_n(\delta_T)$ is realized by strictly positive weights on T. The realization is coefficient-regular.*

Proof. By (14) and the analytic inverse-function theorem, after shrinking neighborhoods, Φ_T is a diffeomorphism $U_T \to V_T$ about zero. We may suppose that all factors

$$H_{T,k}(x) = \frac{A_{T,k}(x) A_{T,k-2}(x)}{A_{T,k-1}(x)^2}$$

are positive on U_T. Choose $\delta_T > 0$ so small that the analytically extended map satisfies

$$\Phi_{\mathrm{sp}}\big([0, \delta_T]^{m-1}\big) \subseteq V_T.$$

For a target (16), set

$$x = \Phi_T^{-1}(\Phi_{\mathrm{sp}}(z)).$$

This inverse lies in the positive orthant. Indeed, (13) gives coordinate by coordinate

$$\vartheta_k^{\mathrm{sp}}(z) = \gamma_k x_{k-1} H_{T,k}(x),$$

whose left side and all factors except x_{k-1} are positive.

It remains to match scale. Since

$$q_1^T = 2s A_{T,1}(x), \qquad p_1 = \tau B_1(z),$$

choose

$$s = \frac{\tau B_1(z)}{2 A_{T,1}(x)} > 0. \tag{19}$$

Then $q_1^T = p_1$ and all the ϑ_k agree. The recursion (6), starting from $q_0 = p_0 = 1$, proves $q_k^T = p_k$ for all k. Thus the characteristic polynomials, and hence the spectra, agree exactly.

Finally, Φ_T is a local diffeomorphism and $\partial q_1^T / \partial s = 2A_{T,1}(x) > 0$. Both $(s, x) \mapsto w$ and $q \mapsto (q_1, \vartheta_2, \dots, \vartheta_m)$ are diffeomorphisms on the positive orthant. The coefficient Jacobian with respect to the m tree weights is therefore nonsingular. □

The chart also records the leading weights. As $z \to 0$,

$$x_{k-1} = \frac{k^2}{k^2-1} z_{k-1}(1 + O(\|z\|)), \qquad w_j = \frac{j}{j+1}\ell_j(1 + O(\|z\|)). \tag{20}$$

The second identity uses $\prod_{k=2}^{j} k^2/(k^2-1) = 2j/(j+1)$ and $s = \ell_1/2\,(1 + O(\|z\|))$.

To pass from a spanning tree to the full graph we need an exact, not merely limiting, perturbation.

Lemma 4.2 (Regular spanning-subgraph lift)**.** *Let H be a connected spanning subgraph of G. Suppose a simple spectrum has an H-realization at which the coefficient map has full rank with respect to $n-1$ selected edge weights of H. Then the same spectrum has a strictly positive, coefficient-regular realization on G.*

Proof. Hold every unselected edge weight of H fixed at its original positive value. Separate the selected weights as $u \in \mathbb{R}^{n-1}$ and the new edge weights as v. Set

$$F(u, v) = (q_1(L_G(u, v)), \dots, q_m(L_G(u, v))).$$

At the original realization $(u^0, 0)$, the derivative $D_u F(u^0, 0)$ is invertible. The implicit-function theorem supplies $u = u(v)$ near zero with $F(u(v), v) = F(u^0, 0)$. Choose every coordinate of v positive and sufficiently small. Continuity gives $u(v) > 0$ and preserves the invertibility of $D_u F$. The new Laplacian has support exactly G and the same characteristic polynomial. □

Proof of Theorem 1.1. For $n = 2$, take $\delta_2 = 1/2$; the only connected graph is K_2, and $w = \lambda_2/2$ is a regular realization, hence has the SSPWL. For $n \geq 3$, choose one prefix order for each labeled tree T on n vertices, and let δ_T be supplied by Proposition 4.1. There are finitely many such trees, so

$$\delta_n = \min\left\{\frac{1}{2}, \min_T \delta_T\right\} > 0.$$

Given a connected G, choose a spanning tree $T \subseteq G$. Proposition 4.1 gives a regular strict realization on T, and Lemma 4.2 lifts it to a regular strict realization on G. The selected tree-weight columns form a nonsingular $m \times m$ minor of the full coefficient Jacobian on G. Since the prescribed spectrum is simple, [8, Theorem 3.8] gives the SSPWL. □

Remark 4.3 (What is and is not uniform). The number δ_n depends only on n, not on G, the overall scale, or the anisotropy among the successive ratios. The proof is qualitative: it does not optimize δ_n. It also requires strict inequalities and a simple target spectrum. Boundary spectra with repeated or zero eigenvalues are outside this chart.

The same regular point also controls the generic algebraic complexity of the tree inverse problem. The following degree is taken over $\mathbb{C}$, with edge labels retained and isolated solutions counted with multiplicity.

Proposition 4.4 (Sharp universal generic-degree bound for trees)**.** *Let T be a labeled tree on n vertices and put $m = n - 1$. On the affine trace hyperplane*

$$H_T = \{w \in \mathbb{C}^m : q_1(w) = 1\},$$

the coefficient map

$$\mathcal{C}_T : H_T \longrightarrow \mathbb{C}^{m-1}, \qquad \mathcal{C}_T(w) = (q_2(w), \dots, q_m(w)),$$

is dominant and generically finite, and

$$\deg \mathcal{C}_T \leq 2 \cdot 3 \cdots m = m!.$$

This bound is sharp: for the labeled star $K_{1,m}$ one has $\deg \mathcal{C}_{K_{1,m}} = m!$.

Proof. Proposition 4.1 supplies a positive point at which the full coefficient Jacobian is nonsingular. Homogeneity separates the radial trace direction, so the derivative of $\mathcal{C}_T$ restricted to $T_w H_T$ is nonsingular after rescaling that point to $q_1 = 1$. Hence $\mathcal{C}_T$ is dominant and generically finite. Eliminate one weight with the linear equation $q_1 = 1$. A generic fiber is then cut out by equations of degrees $2, 3, \dots, m$, so affine Bézout gives the stated upper bound [13].

For the star, every k-edge forest has one nontrivial component of order $k+1$. The rooted-forest formula therefore gives

$$q_k(w) = (k+1)e_k(w_1, \dots, w_m).$$

Together with the fixed value of e_1, a generic target determines the monic polynomial having $w_1, \dots, w_m$ as its roots. Distinct roots give exactly the $m!$ labeled permutations. Thus equality holds. This is a complex generic degree statement; a particular real fiber can be smaller or positive-dimensional on the exceptional algebraic set. □

5. A sharp all-order obstruction for weighted paths

Let P_n have edges $e_i = \{i, i+1\}$, $1 \le i \le n-1$, and let $D \in \mathbb{R}^{n\times(n-1)}$ be the oriented incidence matrix whose ith column is $\mathbf{e}_i - \mathbf{e}_{i+1}$. For positive weights $w_1, \dots, w_{n-1}$ put $W = \operatorname{diag}(w_1, \dots, w_{n-1})$. The nonzero eigenvalues of $L_w = DWD^{\mathsf{T}}$ are the eigenvalues of

$$A_w = W^{1/2} C_{n-1} W^{1/2}, \qquad C_{n-1} = D^{\mathsf{T}} D = \begin{pmatrix} 2 & -1 & & \\ -1 & 2 & \ddots & \\ & \ddots & \ddots & -1 \\ & & -1 & 2 \end{pmatrix}. \tag{21}$$

Proposition 5.1 (Sharp path eigenratio)**.** *For every $n \ge 2$ and every positive weighting of P_n,*

$$\frac{\lambda_n(L_w)}{\lambda_2(L_w)} \ge \frac{1+\cos(\pi/n)}{1-\cos(\pi/n)} = \cot^2\left(\frac{\pi}{2n}\right). \tag{22}$$

Equality holds if and only if $w_1 = \cdots = w_{n-1}$; for $n = 2$ this condition is vacuous.

Proof. Set

$$s_i = \sin\frac{i\pi}{n}, \qquad t_i = (-1)^{i+1} s_i, \qquad 1 \le i \le n-1.$$

The two endpoint eigenpairs of C_{n-1} are

$$C_{n-1} s = \alpha_n s, \qquad C_{n-1} t = \beta_n t,$$

where

$$\alpha_n = 2 - 2\cos\frac{\pi}{n}, \qquad \beta_n = 2 + 2\cos\frac{\pi}{n}.$$

Moreover $s_i^2 = t_i^2$. Apply the Rayleigh quotient of A_w to $x = W^{-1/2}s$ and $y = W^{-1/2}t$. If

$$S_w = s^{\mathsf{T}} W^{-1} s = t^{\mathsf{T}} W^{-1} t, \qquad r_w = \frac{\alpha_n \|s\|^2}{S_w},$$

then

$$\mathcal{Q}_{A_w}(x) = r_w, \qquad \mathcal{Q}_{A_w}(y) = \frac{\beta_n}{\alpha_n} r_w.$$

Thus $\lambda_{\min}(A_w) \le r_w$ and $\lambda_{\max}(A_w) \ge (\beta_n/\alpha_n) r_w$, proving (22).

If equality holds, both Rayleigh inequalities are equalities. In particular, $W^{-1/2}s$ is a smallest-eigenvalue eigenvector of A_w, so for some $r > 0$,

$$W^{1/2} C_{n-1} s = r W^{-1/2} s.$$

Since $C_{n-1}s = \alpha_n s$ and every $s_i > 0$, this says $\alpha_n w_i = r$ for every i. Conversely, if $W = cI$, then $A_w = cC_{n-1}$ and equality follows from the displayed endpoint eigenvalues. □

Remark 5.2 (Classical antecedent). The minimization assertion in Proposition 5.1 is not new as a matrix-scaling theorem. The matrix $C_{n-1}/2$ has unit diagonal and Young's property A after the odd and even edge indices are separated. It is therefore a direct instance of the Forsythe–Straus theorem that Jacobi scaling is optimal, among positive diagonal congruences, for a positive definite matrix with Young's property A [16, Theorem 4]. The proof above is included because the paired sine eigenvectors expose the graph constant and equality case in a few lines. Related path weight optimization appears in [2].

6. A spectral-volume consequence

Put $d = n - 1$ and let

$$\Delta_{d-1} = \left\{ x \in \mathbb{R}^d_{\geq 0} : \sum_{i=1}^{d} x_i = 1 \right\}.$$

Following [14, Section 8], the normalized spectral volume $P(G)$ of a connected n-vertex graph is the relative Euclidean volume in Δ_{d-1} of the closure of its trace-one realizable nonzero spectra, with all coordinate permutations included. Strict and nonnegative support conventions have the same closed spectral region. Indeed, after trace normalization the nonnegative weight simplex is compact, its strict interior is dense, and the ordered spectrum depends continuously on the weights. We write $B(a, b) = \Gamma(a)\Gamma(b)/\Gamma(a + b)$ for Euler's beta function.

Lemma 6.1 (A simplex ratio integral). *Let $d \geq 2$ and $0 < q < 1$. If X is uniform on Δ_{d-1}, then*

$$\Pr\left(\frac{\min_i X_i}{\max_i X_i} \leq q \right) = 1 - \frac{d}{1-q} B\left(\frac{1 + (d-1)q}{1-q}, d \right). \tag{23}$$

Equivalently, with $a = (1+(d-1)q)/(1-q) = 1+dq/(1-q)$, the right side is $1-(d-1)B(a, d-1)$.

Proof. Let $E_1, \ldots, E_d$ be independent mean-one exponential random variables. Then $X_i = E_i/(E_1 + \cdots + E_d)$ is uniform on the simplex, and ratios are unchanged by normalization. The maximum is unique almost surely. Choosing its index and writing its value as t gives

$$\begin{aligned} \Pr(\min_i E_i > q \max_i E_i) &= d \int_0^\infty e^{-t} \left(\int_{qt}^{t} e^{-u}\, du \right)^{d-1} dt \\ &= d \int_0^\infty e^{-[1+q(d-1)]t} (1 - e^{-(1-q)t})^{d-1} dt. \end{aligned}$$

The substitution $z = e^{-(1-q)t}$ gives the beta integral complementary to (23). The equivalent form follows from $(a + d - 1)B(a, d) = (d - 1)B(a, d - 1)$. □

Corollary 6.2 (Beta-function path bound). *For $n \geq 3$, set*

$$q_n = \tan^2\left(\frac{\pi}{2n} \right), \qquad a_n = \frac{1 + (n-2)q_n}{1 - q_n} = 1 + \frac{(n-1)q_n}{1 - q_n}.$$

Then

$$\boxed{P(P_n) \leq U_n := 1 - (n-2)B(a_n, n-2).} \tag{24}$$

Equivalently,

$$U_n = 1 - \frac{n-1}{1-q_n} B(a_n, n-1).$$

Moreover,

$$U_n = \frac{\pi^2}{4n} H_{n-2} + O\left(\frac{(\log n)^2}{n^2} \right) = \frac{\pi^2}{4n} (\log n + \gamma_{\mathrm{E}}) + O\left(\frac{(\log n)^2}{n^2} \right). \tag{25}$$

In particular, $P(P_n) = O((\log n)/n)$.

Proof. A realizable normalized positive spectrum of P_n has $d = n-1$ coordinates and, by Proposition 5.1, satisfies

$$\frac{\min_i \lambda_i}{\max_i \lambda_i} \le q_n.$$

Lemma 6.1 gives the exact volume of this necessary region and hence (24).

For the asymptotic write $a_n = 1+\epsilon_n$. Since

$$q_n = \frac{\pi^2}{4n^2} + O(n^{-4}), \qquad \epsilon_n = \frac{(n-1)q_n}{1-q_n} = \frac{\pi^2}{4n} + O(n^{-2}),$$

we have

$$1 - U_n = (n-2)B(1+\epsilon_n, n-2) = \frac{\Gamma(1+\epsilon_n)\Gamma(n-1)}{\Gamma(n-1+\epsilon_n)}.$$

Using $\psi(n-1) + \gamma_{\mathrm{E}} = H_{n-2}$ and Taylor expansion of $\log\Gamma$ gives

$$\log(1-U_n) = -\epsilon_n H_{n-2} + O(\epsilon_n^2).$$

Since $\epsilon_n H_{n-2} = O((\log n)/n)$, exponentiation proves (25). □

Remark 6.3 (Checks and scope). For $n=2$, $P_2 = K_2$ realizes every one-point positive spectrum, so $P(P_2) = 1$. For $n=3$, $q_3 = 1/3$, $a_3 = 2$, and $U_3 = 1/2$; this is exact by the P_3 characterization in [15]. For $n=4$ the bound is $U_4 = 9/17$, but the exact P_4 region has further constraints [14]. The asymptotic equivalence in (25) belongs to the upper bound U_n, not to $P(P_n)$ itself.

7. Exact lifting to hyperbolic surfaces

We now transfer regular graph charts to closed hyperbolic surfaces of fixed genus. Throughout this section the surface eigenvalues are indexed by

$$0 = \lambda_0(X) < \lambda_1(X) \le \lambda_2(X) \le \cdots.$$

Fix $g \ge 2$ and put

$$N = 2g-2, \qquad m = N-1 = 2g-3. \tag{26}$$

Let T_g be a tree with g vertices of degree one and $g-2$ vertices of degree three. Thus $|V(T_g)| = 2g-2$ and $|E(T_g)| = 2g-3$; for $g=2$ it is a single edge.

Associate a one-holed torus to each leaf and a pair of pants to each trivalent vertex, then glue boundary components according to the edges of T_g. The number of available boundary components is

$$g + 3(g-2) = 4g-6 = 2|E(T_g)|,$$

so all are paired. Every piece has Euler characteristic -1, and hence the closed connected result has

$$\chi = -g - (g-2) = 2-2g.$$

It is a surface of genus g.

This decomposition is adapted to Burger's weighted graph. If Y_v is a piece, its graph vertex mass in the normalization of [3] is

$$\mathfrak{m}(v) = -\chi(Y_v) = 1$$

for both a pair of pants and a one-holed torus. Equivalently, every piece has area 2π. The limiting operator is therefore the ordinary weighted combinatorial Laplacian with unit vertex masses.

For $w \in \mathbb{R}^m_{>0}$ let

$$L_{T_g}(w) = \sum_{e=uv\in E(T_g)} w_e(\mathbf{e}_u - \mathbf{e}_v)(\mathbf{e}_u - \mathbf{e}_v)^{\mathsf{T}},$$

and write its positive eigenvalues as

$$0 < \mu_1(w) \le \cdots \le \mu_m(w), \qquad \Lambda_{T_g}(w) = (\mu_1(w), \dots, \mu_m(w)).$$

Choose the remaining Fenchel–Nielsen data once and for all. Inside every one-holed torus, fix the length and twist of a nonseparating curve, and fix the twists along the separating curves represented by T_g. For $t > 0$ and $w > 0$, let $X(t, w)$ be the closed curvature--1 surface whose separating cuff associated to e has length

$$\ell_e(t, w) = tw_e. \tag{27}$$

These are all $3g - 3$ length coordinates and all $3g - 3$ twist coordinates of a pants decomposition. Thus $X(t, w)$ is a well-defined marked hyperbolic surface, depending real-analytically on (t, w) for $t > 0$.

Let $A(t, w)$ be the multicurve formed by the m separating cuffs. We first record the uniform geometric input needed in Burger's comparison theorem. Following [3], write

$$L(A) = \min\{2\,\mathrm{arsinh}(1),\ \ell(\gamma) : \gamma \text{ is a closed geodesic disjoint from } A \text{ and not peripheral to } A\}, \tag{28}$$

with the first entry retained when the second set is empty, and put $l(A) = \max_{\alpha \in A} \ell(\alpha)$.

Lemma 7.1 (Uniform nonpinching scale)**.** *For every compact $C \Subset \mathbb{R}^m_{>0}$ there are $t_C, c_C > 0$ such that, for $0 < t < t_C$ and $w \in C$, Burger's nonpinching length scale satisfies*

$$L(A(t, w)) \ge c_C.$$

Consequently Burger's cusp parameter obeys

$$\epsilon(t, w) := \frac{l(A(t, w))}{L(A(t, w))} \le \frac{t \max_{w \in C, e} w_e}{c_C} = O_C(t). \tag{29}$$

Proof. The assertion is immediate on every parameter subinterval bounded away from $t = 0$, by compactness of Fenchel–Nielsen coordinates. Consider therefore a sequence with $t \downarrow 0$. A closed geodesic disjoint from $A(t, w)$ lies in one complementary piece. A pair of pants contains no nonperipheral essential simple closed curve. Any closed geodesic shorter than $2\,\mathrm{arsinh}(1)$ is simple, so a sequence of closed geodesics with length tending to zero is eventually simple. A one-holed torus has fixed positive internal length and fixed twist coordinates, while its boundary length tends to zero in a bounded interval. The fixed positive internal length excludes internal nodal strata. These tori therefore remain in a compact subset of the augmented one-holed-torus moduli space, and their nonperipheral systoles have a uniform positive lower bound. A curve meeting a component of $A(t, w)$ cannot become short, by the collar lemma. These observations give the required lower bound for the nonpinching scale; see, for example, [4, Chapters 3–4]. Since $l(A(t, w)) = t \max_e w_e$, (29) follows. □

Lemma 7.2 (Uniform Burger limit)**.** *For every compact $C \Subset \mathbb{R}^m_{>0}$,*

$$\lim_{t \downarrow 0} \sup_{w \in C} \max_{1 \le j \le m} \left| \frac{2\pi^2}{t} \lambda_j(X(t, w)) - \mu_j(w) \right| = 0. \tag{30}$$

Proof. For the partition by the m separating cuffs, Burger's graph is T_g, every vertex mass is one, and the conductance vector is tw. Its positive eigenvalues, after discarding the common zero mode, are $t\mu_1(w), \dots, t\mu_m(w)$. Lemma 7.1 gives $\epsilon = O_C(t)$ uniformly. Hence, for t small enough that Burger's condition $\epsilon < \alpha_1$ holds throughout C, the uniform comparison [3, Theorem 1.1] gives, in the standard combinatorial normalization,

$$\frac{\lambda_j(X(t, w))}{t\mu_j(w)} = \frac{1}{2\pi^2}(1 + o(1)).$$

The error is uniform in a fixed topological cusp neighborhood; it is controlled there by a constant times $\sqrt{\epsilon} + \epsilon|\log\epsilon|$. The functions $\mu_j(w)$ are bounded on C, so the multiplicative estimate implies the absolute convergence (30). In addition, Otal–Rosas gives the pointwise bound $\lambda_{m+1} = \lambda_{2g-2} > 1/4$, identifying these as exactly the collapsing positive modes. □

Remark 7.3 (The normalization constant). The factor $2\pi^2$ must be retained when $L_{T_g}(w)$ denotes the standard combinatorial Laplacian. Burger's vertex mass is one here, whereas the hyperbolic area of each piece is 2π. Equivalently, the graph quadratic form $\pi^{-1}\sum_e w_e(F_u - F_v)^2$ with scalar product $2\pi\sum_v F_v G_v$ has operator $(2\pi^2)^{-1}L_{T_g}(w)$. Mixing these two conventions would change the prescribed scale.

The next theorem is the exact-lifting step. It uses only continuous dependence of ordered eigenvalues and Brouwer degree, not differentiability of the surface spectral map at the nodal limit.

Theorem 7.4 (Regular graph charts give exact surface spectra)**.** *Suppose $\rho = (\rho_1, \dots, \rho_m)$, with $0 < \rho_1 < \cdots < \rho_m$, has a strict regular realization on T_g: there is $w^0 > 0$ such that*

$$\Lambda_{T_g}(w^0) = \rho, \qquad \det D\Lambda_{T_g}(w^0) \neq 0.$$

Then there are a neighborhood V_ρ of ρ and $s_\rho > 0$ such that, for every $\eta \in V_\rho$ and every $0 < s < s_\rho$, some closed hyperbolic surface X of genus g satisfies

$$(\lambda_1(X), \dots, \lambda_m(X)) = s\eta. \tag{31}$$

The threshold may be chosen so that $s\eta_m < 1/4$ throughout this set.

Proof. The inverse-function theorem gives a connected neighborhood U_0 of w^0 on which Λ_{T_g} is a diffeomorphism onto its image. Shrink U_0 inside the simple-spectrum locus, and choose $r > 0$ so that

$$\overline{B_{2r}(\rho)} \subset \Lambda_{T_g}(U_0) \cap \{0 < \eta_1 < \cdots < \eta_m\}.$$

Set

$$D = (\Lambda_{T_g}|_{U_0})^{-1}(B_{2r}(\rho)).$$

After shrinking U_0, we have $\overline{D} \Subset \mathbb{R}^m_{>0}$, and Λ_{T_g} maps D diffeomorphically onto $B_{2r}(\rho)$.

For $t > 0$ define the continuous map

$$\Phi_t(w) = \frac{2\pi^2}{t}(\lambda_1(X(t,w)), \dots, \lambda_m(X(t,w))), \qquad w \in \overline{D}.$$

Continuity follows from the Fenchel–Nielsen construction of $X(t,w)$ and the min–max characterization of ordered Laplace eigenvalues. Lemma 7.2 gives

$$\sup_{w\in\overline{D}} \|\Phi_t(w) - \Lambda_{T_g}(w)\| \longrightarrow 0. \tag{32}$$

Put $V_\rho = B_r(\rho)$. For $\eta \in V_\rho$ and $w \in \partial D$, $\|\Lambda_{T_g}(w) - \eta\| \geq r$. Consequently, for all sufficiently small t, uniformly in $\eta \in V_\rho$, the straight-line homotopy from Λ_{T_g} to Φ_t avoids η on ∂D. Homotopy invariance of Brouwer degree gives

$$\deg(\Phi_t, D, \eta) = \deg(\Lambda_{T_g}, D, \eta) = \operatorname{sgn}\det D\Lambda_{T_g}(w^0) \neq 0.$$

For the last equality, the graph map has one preimage of η in D, and the sign of its nonzero Jacobian is constant on the connected set D. Thus $\Phi_t(w(t,\eta)) = \eta$ for some $w(t,\eta) \in D$, or

$$\lambda_j(X(t, w(t,\eta))) = \frac{t}{2\pi^2}\eta_j.$$

Taking $t = 2\pi^2 s$ proves (31); shrink s_ρ to obtain the final inequality. □

The graph base point may depend on ρ without causing a uniformity problem. For a fixed target it is chosen once, and the same degree domain works at every sufficiently small scale; the realizing weight is allowed to vary with the scale. The argument supplies existence at each scale, not a continuous selection of realizing weights or surfaces.

Corollary 7.5 (Compact-uniform exact lift)**.** *Let $\Omega(T_g)$ be the set of simple spectra admitting a strict regular realization on T_g. For every compact $K \Subset \Omega(T_g)$ there is $s_{g,K} > 0$ such that, for every $\rho \in K$ and $0 < s < s_{g,K}$, some closed hyperbolic surface X of genus g satisfies*

$$(\lambda_1(X), \dots, \lambda_{2g-3}(X)) = s\rho.$$

Proof. The neighborhoods V_ρ from Theorem 7.4 cover K. Take a finite subcover and the minimum of the associated positive thresholds. □

8. Full-index prescription in genus g

Proof of Theorem 1.6. Take

$$\varepsilon_g = \delta_{2g-2}$$

from Theorem 1.1. The tree T_g is connected on $2g-2$ vertices, so every $\rho \in \mathcal{L}_{2g-2}(\varepsilon_g)$ has a strict regular realization on T_g. Theorem 7.4 gives the exact equality for every sufficiently small s. The theorem of Otal–Rosas [22] states that

$$\lambda_{2g-2}(X) > \frac{1}{4} \tag{33}$$

for every closed hyperbolic surface of genus g. After reducing the threshold so that $s\rho_{2g-3} < 1/4$, the prescribed list is therefore precisely the full collection of eigenvalues below $1/4$, and the next eigenvalue satisfies the asserted strict $1/4$ bound. □

Corollary 8.1 (A compact lacunary region)**.** *If $K \Subset \mathcal{L}_{2g-2}(\varepsilon_g)$, there is one $s_{g,K} > 0$ such that every $s\rho$, with $\rho \in K$ and $0 < s < s_{g,K}$, is exactly the first $2g-3$ positive-eigenvalue list of a closed genus-g hyperbolic surface.*

Proof. Combine Theorem 1.1 with Corollary 7.5, reducing the threshold using (33). □

This gives an exact general-genus near-origin region for [14, Question 1]. This lacunary corollary is a partial answer: it neither characterizes the entire cube $[0, 1/4]^{2g-3}$ nor treats lists with repeated entries. The exact graph-to-surface degree mechanism is classical: it is the essential-family construction of Colbois–Colin de Verdière [10, Lemma V.1 and Section VI], used there with a complete graph to create high multiplicity. The ingredient specific to the present paper is the universal regular lacunary chart. Choosing it on the genus-compatible tree T_g reaches the full admissible length $2g-3$.

9. The complexity-one stable-graph spectral link

We now compactify the preceding single graph chart. A *complexity-one stable genus-g pseudograph* is a connected pseudograph Γ, with loops and parallel edges allowed, having $N = 2g-2$ vertices, arithmetic genus

$$b_1(\Gamma) + \sum_v h(v) = g, \tag{34}$$

and a genus label $h(v) \geq 0$ at every vertex such that

$$2h(v) - 2 + \deg_\Gamma(v) = 1. \tag{35}$$

Loops count twice in the degree. Thus a vertex represents either a pair of pants, $(h, \deg) = (0, 3)$, or a one-holed torus, $(h, \deg) = (1, 1)$. There are only finitely many isomorphism types; denote their collection by $\mathfrak{G}_g$.

Loops remain part of the topology but contribute no energy to functions constant on a component. Combine the conductances of all parallel nonloop cuffs joining the same two vertices, and call the resulting set of distinct pairs $E_{\mathrm{eff}}(\Gamma)$. On the compact trace-one simplex put

$$K_\Gamma = \left\{ w \in \mathbb{R}_{\geq 0}^{E_{\mathrm{eff}}(\Gamma)} : 2\sum_e w_e = 1 \right\}.$$

If $0 = \mu_1 \leq \cdots \leq \mu_N$ are the eigenvalues of $L_\Gamma(w)$, including any extra zeros caused by the positive support disconnecting, set

$$\psi_\Gamma(w) = (\mu_2, \ldots, \mu_N).$$

We shall use the comparison theorem also when the cuff lengths have arbitrarily different orders. The following formulation isolates the needed uniformity.

Lemma 9.1 (Uniform projectivized Burger comparison)**.** *Let X_k converge in the augmented moduli space after pinching a nonempty full Deligne–Mumford multicurve A_k, and suppose that the complement has r components. Let G_k be Burger's weighted graph, with vertex masses $-\chi(Y_{v,k})$ and cuff-length conductances. Then, after passing to the convergent augmented chart,*

$$L(A_k) \geq c > 0, \qquad \epsilon_k = \frac{l(A_k)}{L(A_k)} \longrightarrow 0,$$

and

$$\liminf_{k\to\infty} \lambda_r(X_k) > 0. \tag{36}$$

If $r \geq 2$ and $0 < \nu_{1,k} \leq \cdots \leq \nu_{r-1,k}$ are the positive graph eigenvalues, then

$$\max_{1\leq j\leq r-1} \left| \frac{2\pi^2 \lambda_j(X_k)}{\nu_{j,k}} - 1 \right| \longrightarrow 0. \tag{37}$$

Thus exactly $r-1$ positive surface eigenvalues tend to zero. The estimate is independent of the ratios among the pinching cuff lengths. Consequently, when $r \geq 2$ and the graph spectral trace is normalized, the projectivized surface and graph spectra have the same subsequential limits, including limits on weight-simplex faces.

Proof. Fullness means that, on each component of the limiting noded surface, no additional essential geodesic disjoint from A_k tends to zero. Compactness in the chosen augmented chart and the definition (28) therefore give $L(A_k) \geq c > 0$, whereas $l(A_k) \to 0$. Burger's multiplicative estimate [3, Theorem 1.1] depends only on the topology and ϵ_k; it does not involve the smallest-to-largest conductance ratio. It applies simultaneously to the $r-1$ positive graph modes and gives (37) when $r \geq 2$. Because every cuff in A_k shrinks, all these graph eigenvalues tend to zero. On the orthogonal complement of the componentwise constants, the limiting components have a positive Poincaré threshold; compactness of the augmented chart makes that threshold uniform. The min–max principle, equivalently the noncollapsing part of Burger's comparison, gives (36). Hence there are exactly $r-1$ collapsing positive modes. Dividing the collapsing spectral lists by their sums cancels the common constant and proves the last assertion. □

Theorem 9.2 (Exact maximal-cusp tangent link)**.** *For every $g \geq 2$,*

$$\boxed{\mathcal{T}_g = \bigcup_{\Gamma\in\mathfrak{G}_g} \psi_\Gamma(K_\Gamma).} \tag{38}$$

In particular, $\mathcal{T}_g$ is compact and semialgebraic. Equivalently, without fixing trace, the right side consists of

$$\frac{(\mu_2(L_\Gamma(w)), \ldots, \mu_N(L_\Gamma(w)))}{\operatorname{tr} L_\Gamma(w)}, \qquad w \geq 0, \quad \operatorname{tr} L_\Gamma(w) > 0.$$

The zero conductances in (38) *record multiscale pinching; they cannot in general be discarded.*

Proof. Suppose $\lambda_m(X_k) \to 0$. The sequence leaves every compact part of moduli space, so after passing to a Deligne–Mumford subsequence it has a nonempty full pinching multicurve A_k. If its complement has r components, Burger's cusp-partition theorem and Lemma 9.1 give precisely $r-1$ collapsing graph modes. Thus $r-1 \geq m$. On the other hand, stability gives

$$\sum_{v=1}^{r} -\chi(Y_v) = 2g-2 = N,$$

and every summand is a positive integer. Hence $r \leq N = m+1$, so equality holds throughout: $r = N$, there are exactly m collapsing positive modes, and every component has mass one. Each component is therefore a pair of pants or a one-holed torus, and the stable dual type belongs to $\mathfrak{G}_g$. Otal–Rosas supplies independently $\lambda_{m+1}(X_k) > 1/4$.

Pass to a subsequence on which the finite graph type is fixed. After aggregating parallel cuffs and normalizing total conductance, compactness of K_Γ gives $w_k \to w$. Burger's multiplicative comparison, with the common normalization cancelled, remains valid as the normalized weights approach a face by Lemma 9.1. It yields $p(X_k) \to \psi_\Gamma(w)$. Vanishing coordinates of w retain the extra zero graph eigenvalues and therefore cover multiscale limits. This proves one inclusion.

Conversely, fix (Γ, w) on the right. Approximate every zero conductance by trace-one vectors $w^{(a)} > 0$ with $w^{(a)} \to w$, split an effective conductance positively among parallel cuffs, and choose positive lengths for loop cuffs. Realize the vertices by their complexity-one hyperbolic pieces, freeze the remaining Fenchel–Nielsen data, and multiply all cuff lengths by t. For each a, Lemma 9.1 gives the direction $\psi_\Gamma(w^{(a)})$ as $t \downarrow 0$. Choose $t_a > 0$ so that the projectivized error is below $1/a$ and $\lambda_m < 1/a$. Then the diagonal sequence converges to $\psi_\Gamma(w)$ while $\lambda_m \to 0$.

Finally, compactness follows from the finite union in (38). Semialgebraicity follows by expressing the ordered eigenvalues as ordered roots of the characteristic polynomial and applying Tarski–Seidenberg to each compact weight simplex. □

Corollary 9.3 (Off-link exclusion)**.** *If $p = (p_1, \dots, p_m)$ satisfies $0 < p_1 \leq \cdots \leq p_m$ and $\sum_j p_j = 1$, and $p \notin \mathcal{T}_g$, then there is $s_p > 0$ such that sp does not occur as the first m positive-eigenvalue list of a genus-g closed hyperbolic surface for any $0 < s < s_p$.*

Proof. Otherwise choose $s_k \downarrow 0$ and surfaces whose first m positive-eigenvalue lists are $s_k p$. Their normalized directions equal p and their mth eigenvalues tend to zero, contradicting the definition of $\mathcal{T}_g$. □

Corollary 9.4 (Compact-uniform off-link exclusion)**.** *Let K be a compact subset of*

$$\left\{ 0 < p_1 \leq \cdots \leq p_m : \sum_j p_j = 1 \right\} \setminus \mathcal{T}_g.$$

There is one $s_K > 0$ such that no sp, with $p \in K$ and $0 < s < s_K$, occurs as the first m positive-eigenvalue list of a closed genus-g hyperbolic surface.

Proof. Otherwise there are $p_k \in K$, $s_k \downarrow 0$, and realizing surfaces. After taking a subsequence, $p_k \to p \in K$. Their mth eigenvalues tend to zero, so the definition of the tangent link gives $p \in \mathcal{T}_g$, a contradiction. □

10. Generic exactness and the singular skeleton

We allow every relative-open face of every K_Γ. A point w on such a face is *normalized regular* if $\psi_\Gamma(w)$ has positive, pairwise distinct coordinates and the derivative along the face has rank

$m-1$. Define

$$\mathcal{U}_g = \{p \in \mathcal{T}_g : p \text{ has a normalized-regular preimage on some face}\}, \qquad \mathfrak{B}_g = \mathcal{T}_g \setminus \mathcal{U}_g. \tag{39}$$

This definition is deliberately existential. A spectral value can have a critical preimage and a regular preimage simultaneously, so the image of all critical weights is not the boundary and is not the correct skeleton.

We first extend Theorem 7.4 from the chosen tree to a stable graph. Split parallel effective weights in fixed positive proportions and freeze all loop and unused cuff variables.

Lemma 10.1 (Uniform convergence in a stable chart)**.** *Fix $\Gamma \in \mathfrak{G}_g$, a positive splitting of each effective conductance among its parallel cuffs, positive rescaled lengths for all loop cuffs, and compact thick and twist data on the complementary pieces. If $C \Subset \mathbb{R}_{>0}^{E_{\mathrm{eff}}(\Gamma)}$, let $X_\Gamma(t,w)$ be the surface obtained by multiplying all nodal cuff lengths by t. Writing $\Lambda_\Gamma(w) = (\mu_2(w), \dots, \mu_N(w))$, one has*

$$\sup_{w \in C} \left\| \frac{2\pi^2}{t} \Lambda_g(X_\Gamma(t,w)) - \Lambda_\Gamma(w) \right\| \longrightarrow 0 \qquad (t \downarrow 0). \tag{40}$$

The same conclusion is uniform when the fixed auxiliary data range over a compact subset of the positive parameter locus; in particular, all splitting proportions and rescaled loop-cuff lengths then stay uniformly away from zero.

Proof. All complementary components have complexity one and hence Burger mass one. The chosen positive splittings sum to the prescribed effective conductances, whereas loop cuffs contribute no graph energy. Compactness of C and of the auxiliary data gives a uniform lower bound for $L(A)$ by the argument of Lemma 7.1; the cusp parameter is therefore $O(t)$. Burger's topology-uniform multiplicative comparison then gives (40). The final assertion follows from the same estimates on the product compact set. □

Proposition 10.2 (Stable-graph regular transfer)**.** *Let $\Gamma \in \mathfrak{G}_g$ have a strict effective weighting w^0 with simple positive spectrum ρ. If the unnormalized spectral derivative has rank m, then there are a neighborhood V_ρ and $s_\rho > 0$ such that every $s\eta$, $\eta \in V_\rho$ and $0 < s < s_\rho$, is exactly the first m positive-eigenvalue list of a genus-g closed hyperbolic surface.*

Proof. Choose an invertible $m \times m$ spectral minor and vary only those cuff weights on a small affine slice, keeping all other positive weights fixed. On a compact subset of that slice, Lemma 10.1 makes the scaled surface eigenvalue map converge uniformly to the graph map. The latter has local Brouwer degree ± 1. Choosing a degree domain and a smaller target ball exactly as in Theorem 7.4, uniform convergence makes the straight homotopy boundary-avoiding. Nonzero degree gives the desired preimage. Otal–Rosas separates the next surface eigenvalue from the collapsing cluster. □

Theorem 10.3 (Generic exact maximal-cusp lifting)**.** *For every $g \geq 2$, $\mathcal{U}_g$ is relatively open and semialgebraic in the strict ordered simplex (rather than merely relatively open in $\mathcal{T}_g$), and $\mathfrak{B}_g$ is compact semialgebraic with*

$$\dim \mathfrak{B}_g \leq m-2, \qquad \mathcal{H}^{m-1}(\mathfrak{B}_g) = 0. \tag{41}$$

Here $\dim \varnothing = -\infty$; in particular, $\mathfrak{B}_2 = \varnothing$. For every $p \in \mathcal{U}_g$ there are a relative neighborhood V_p and $s_p > 0$ such that every $s\eta$, $\eta \in V_p$ and $0 < s < s_p$, is exactly realized by a genus-g closed hyperbolic surface. Consequently almost every direction in $\mathcal{T}_g$, with respect to ambient $\mathcal{H}^{m-1}$ measure in the trace-one hyperplane, has an exact all-small-scale ray lift.

Proof. For each relative-open face F of K_Γ, restrict the normalized map to its simple-spectrum locus and denote it by $\psi_{\Gamma,F}$. Let $\operatorname{Crit} \psi_{\Gamma,F}$ be the set where its derivative has rank less than

$m-1$; this is the whole source when $\dim F < m-1$. Let Δ_{sing} be the union of the coordinate and collision walls in the ordered trace-one simplex. The existential definition of regularity gives the explicit containment

$$\mathfrak{B}_g \subseteq \Delta_{\mathrm{sing}} \cup \bigcup_{\Gamma\in\mathfrak{G}_g} \bigcup_{F\preceq K_\Gamma} \psi_{\Gamma,F}(\operatorname{Crit}\psi_{\Gamma,F}). \tag{42}$$

Semialgebraic Sard theory [18] puts every critical value set on the right in dimension at most $m-2$; faces of smaller dimension satisfy the same bound directly. The walls also have dimension at most $m-2$. Since the union is finite, (41) follows.

Characteristic coefficients, Jacobian minors, and quantifier elimination show that $\mathcal{T}_g$, $\mathcal{U}_g$, and $\mathfrak{B}_g$ are semialgebraic. At a normalized submersion, the submersion theorem supplies a neighborhood in the ambient trace-one hyperplane, all of whose points retain regular preimages. Hence $\mathcal{U}_g$ is open in the strict ordered simplex. Its complement in the compact set $\mathcal{T}_g$ is therefore closed, proving compactness of $\mathfrak{B}_g$. For $g=2$, both the tangent link and the trace-one target are the singleton (1), whose zero-dimensional map is regular, so the skeleton is empty. Finally, the lacunary tree chart gives a full-dimensional subset of $\mathcal{T}_g$; thus the ambient-measure form of "almost every" is nonvacuous.

Fix a regular preimage. Homogeneity adds the radial direction and turns normalized rank $m-1$ into unnormalized rank m. Because $p_1 > 0$, $\ker L_\Gamma(w) = \mathbb{R}\mathbf{1}$; equivalently, the positive effective support is a connected spanning subgraph. If some allowed graph weights vanish, the regular spanning-subgraph lift Lemma 4.2 makes every new effective adjacency positive while preserving the spectrum and rank. Parallel weights can then be split and loops assigned arbitrary positive lengths. Proposition 10.2 gives the claimed exact neighborhood and ray. □

Corollary 10.4 (Compact-uniform wedges)**.** *If $K \Subset \mathcal{U}_g$, then one $s_K > 0$ works for every $p \in K$ and $0 < s < s_K$.*

Proof. Take a finite subcover of K by the neighborhoods in Theorem 10.3 and the minimum of their scale thresholds. □

The following elementary inclusion is useful when reading the skeleton. If $F : K \to \mathbb{R}^d$ is C^1 on a compact manifold with corners, and $\operatorname{Crit} F$ denotes the interior points where DF is not onto, then

$$\partial F(K) \subseteq F(\partial K \cup \operatorname{Crit} F). \tag{43}$$

Indeed, an interior preimage at which DF is onto makes the image point interior by the submersion theorem. The converse to (43) is false: the existence of a boundary or critical preimage does not make its value a boundary point.

11. Uniformly conditioned clusters and finite-prefix nonrigidity

The regular part of the maximal-cusp link contains a full-dimensional region whose internal condition number is bounded independently of the genus. The input is the finite-size strict form of the bipartite Ramanujan theorem.

Theorem 11.1 (An epsilon-free Ramanujan wedge)**.** *For every $g \geq 2$, with $m = 2g-3$, there are a nonempty open set*

$$V_g \subset \{0 < \eta_1 < \cdots < \eta_m\}$$

and $s_g > 0$ such that every $s\eta$, $\eta \in V_g$ and $0 < s < s_g$, is exactly the first m positive-eigenvalue list of a closed genus-g hyperbolic surface, with

$$\frac{\eta_m}{\eta_1} < 18 + 12\sqrt{2}. \tag{44}$$

Otal–Rosas gives $\lambda_{m+1} = \lambda_{2g-2} > 1/4$ for every realizing surface. The threshold is reduced only to ensure $s\eta_m < 1/4$; hence the prescribed list is the entire spectrum below $1/4$.

Proof. Write $N = 2g - 2 = 2d$. Theorem 1.1 of Marcus–Spielman–Srivastava [20], applied with degree three and bipartition size $d + d$, gives a cubic bipartite multigraph Γ on N vertices whose nontrivial adjacency eigenvalues α satisfy the strict finite-size estimate

$$|\alpha| < 2\sqrt{2}.$$

The eigenvalue 3 is therefore simple, so Γ is connected. Moreover

$$|E(\Gamma)| = \frac{3N}{2} = 3g - 3, \qquad b_1(\Gamma) = |E| - |V| + 1 = g.$$

Thus Γ is an all-pants complexity-one stable genus-g pseudograph; parallel edges are legitimate cuff edges.

Give every cuff unit conductance and aggregate the multiplicities between each vertex pair. The resulting effective weighted Laplacian is $L = 3I - A$. The trivial bipartite adjacency eigenvalue -3 gives $\mu_{\max} = 6$, while

$$\mu_{\min} > 3 - 2\sqrt{2}.$$

Consequently

$$\frac{\mu_{\max}}{\mu_{\min}} < \frac{6}{3 - 2\sqrt{2}} = 18 + 12\sqrt{2}. \tag{45}$$

Let $\widehat{\Gamma}$ be the connected effective support after parallel edges are aggregated. Theorem 1.1 supplies a simple, coefficient-regular positive weighting on $\widehat{\Gamma}$. Hence some m-row coefficient-Jacobian minor and the spectral discriminant are nonzero polynomials on its weight space. Their simultaneous nonvanishing is dense. We may therefore perturb the aggregated unit-conductance vector arbitrarily slightly to a positive weighting w^* having simple spectrum and unnormalized spectral rank m. The strict margin in (45) preserves (44). Split each effective conductance into fixed positive conductances on its parallel cuffs and apply Proposition 10.2. Shrinking the resulting neighborhood preserves the strict ratio, and Otal–Rosas supplies the last assertions. For $g = 2$ the graph is the two-vertex triple-edge graph; its single positive eigenvalue has condition number one and the same argument is immediate. □

Corollary 11.2 (A maximal cluster with arbitrary outside gap)**.** *For every $g \geq 2$ and every $M > 0$, there is a closed genus-g hyperbolic surface such that*

$$\frac{\lambda_{2g-3}}{\lambda_1} < 18 + 12\sqrt{2}, \qquad \frac{\lambda_{2g-2}}{\lambda_{2g-3}} > M.$$

Proof. Choose $\eta \in V_g$ and then $0 < s < \min\{s_g, 1/(4M\eta_m)\}$. Theorem 11.1 gives $\lambda_m = s\eta_m$ and $\lambda_{m+1} > 1/4$. □

We next make explicit the parameter uniformity behind finite-prefix nonrigidity. The statement uses normalized twist angles and rescaled cuff weights in a local augmented Fenchel–Nielsen chart; these stay in a compact set as the common pinching scale tends to zero.

Proposition 11.3 (Parametric stable-graph transfer)**.** *Let Γ be an all-pants complexity-one stable genus-g graph and let $w^0 > 0$ be a simple, unnormalized-regular effective weighting with spectrum ρ. There are a neighborhood V_ρ, a number $s_\rho > 0$, and a compact parameter cell Q of dimension*

$$(6g - 6) - (2g - 3) = 4g - 3$$

with nonempty interior such that, for every $\eta \in V_\rho$, $0 < s < s_\rho$, and $q \in Q$, some surface with complementary parameter q has

$$(\lambda_1, \ldots, \lambda_{2g-3}) = s\eta, \qquad \lambda_{2g-2} > \frac{1}{4}.$$

Proof. Choose an invertible $m \times m$ spectral minor at w^0 and a degree domain D in those m effective cuff variables, exactly as in Proposition 10.2. The remaining rescaled cuff weights and normalized twist angles give $4g-3$ complementary Fenchel–Nielsen parameters. Let them range in a sufficiently small compact cell Q in one augmented local chart.

The unselected rescaled cuff weights can change the limiting graph map. Denote that map in the selected variables by $\Lambda_{\Gamma,q}$; twist coordinates do not enter it. At the base value q^0, the chosen degree domain has degree ± 1 over a target ball. After shrinking Q and the ball, the straight-line homotopy from Λ_{Γ,q^0} to $\Lambda_{\Gamma,q}$ avoids the target on ∂D, uniformly in $q \in Q$.

The complementary pieces and all rescaled data range over a compact family on $\overline{D} \times Q$. The parameter-uniform form of Lemma 10.1 consequently applies there. If $\Phi_{t,q}$ is the scaled surface spectral map in the selected weights, then

$$\sup_{(w,q)\in\overline{D}\times Q} \|\Phi_{t,q}(w) - \Lambda_{\Gamma,q}(w)\| \longrightarrow 0 \qquad (t \downarrow 0).$$

The displayed uniform convergence makes the surface-to-graph homotopy boundary-avoiding simultaneously for every $q \in Q$. Thus

$$\deg(\Phi_{t,q}, D, \eta) = \deg(\Lambda_{\Gamma,q}, D, \eta) = \deg(\Lambda_{\Gamma,q^0}, D, \eta) = \pm 1$$

for every $q \in Q$, every target in the smaller ball, and every sufficiently small t. Taking $t = 2\pi^2 s$ proves the claim, and Otal–Rosas separates the next eigenvalue. □

Theorem 11.4 (Continuum maximal-prefix nonrigidity)**.** *The sets V_g and constants s_g in Theorem 11.1 may be chosen so that every $s\eta$, $\eta \in V_g$ and $0 < s < s_g$, is shared by continuum many pairwise nonisometric closed genus-g hyperbolic surfaces. All these surfaces have the same entire sub-$1/4$ spectrum, with multiplicity.*

For each fixed s, almost every target in sV_g admits a regular realization in a marked Teichmüller chart. Near such a realization its marked isolevel is a real-analytic manifold of dimension $4g-3$; after passage to moduli the corresponding local object is an orbifold quotient.

Proof. Apply Proposition 11.3 to the regular Ramanujan weighting in the proof of Theorem 11.1. For every $q \in Q$ the degree argument supplies at least one selected weight vector. The resulting set of marked surfaces projects onto the continuum set Q. Each unmarked isometry class has at most countably many markings, since the mapping class group is countable. Hence the family contains continuum many unmarked isometry classes. The $1/4$ statement follows from Otal–Rosas and the choice $s\eta_m < 1/4$.

On the locus where the prescribed eigenvalues are distinct and separated from the next eigenvalue, the spectral map on marked Teichmüller charts is real analytic. Every preimage of a target in sV_g lies in this locus: the target entries are distinct and below $1/4$, while Otal–Rosas puts $\lambda_{m+1} > 1/4$. A countable atlas covers this locus. The union of the critical-value sets of its chart maps is null by Sard's theorem, while the parametric degree argument shows that every point of sV_g has a preimage. Thus almost every target has a regular marked realization, and the analytic regular-level theorem gives dimension $(6g-6) - (2g-3) = 4g-3$. Quotienting by the mapping class action gives the stated local orbifold interpretation. This is finite-prefix nonrigidity, not full isospectrality. It occurs in the degenerating thin regime, in contrast with the finite-spectral-determination theorem of Buser–Courtois on each fixed thick part [5]. □

Theorem 11.5 (Superexponentially many thin chambers)**.** *As $g \to \infty$, there are*

$$\textit{at least } \exp\{g \log g - O(g)\}$$

pairwise nonisomorphic all-pants dual-graph chambers which share one exact maximal sub-$1/4$ prefix. More precisely, for every sufficiently large g there are a simple lacunary list ρ_g and $s_g > 0$

such that every $s\rho_g$, $0 < s < s_g$, is realized by continuum many surfaces in each of that many chambers.

Proof. Put $N = 2g - 2$ and choose one simple list ρ_g in the universal cone $\mathcal{L}_N(\delta_N)$. Every connected simple cubic graph on N vertices has a strict regular realization of ρ_g by Theorem 1.1. Such a graph has $3g - 3$ edges and first Betti number g, so it is an all-pants dual graph. Apply Proposition 11.3 separately to every graph chart. There are only finitely many types for fixed g, so the minimum of their positive exact-lift thresholds is positive. This repeated parametric-degree argument, rather than an appeal to a single Ramanujan chart, gives continuum many realizations inside every chamber.

The configuration-model enumeration of labeled simple cubic graphs is [1]

$$(e^{-2} + o(1))\frac{(3N)!}{(3N/2)!\,2^{3N/2}\,6^N}.$$

Asymptotically almost every cubic graph is connected; for example this follows from the stronger Hamiltonicity theorem [25]. Dividing by $N!$ gives a lower bound on the number of unlabelled types whose logarithm is

$$\frac{N}{2}\log N + O(N) = g\log g + O(g).$$

Finally, after decreasing the common scale threshold, the canonical short multicurve recovers the dual graph. Nonisomorphic dual graphs therefore belong to distinct isometry classes of thin chambers. □

12. Essential collision strata

At a repeated eigenvalue, ordered eigenvalues are not differentiable and the preceding Jacobian degree cannot be used. The appropriate object is the low-energy quadratic form. Let $V_0 = \mathbf{1}^\perp \subset \mathbb{R}^N$. An *all-pants dual graph* is a connected cubic pseudograph representing a decomposition into genus-zero pairs of pants; its loop cuffs have zero graph energy.

Lemma 12.1 (Uniform effective-form convergence)**.** *Fix a strictly weighted genus-g all-pants dual graph Γ and a compact set C of positive effective weights, with the positive parallel-edge splittings, loop coefficients, and compact auxiliary data fixed as in Lemma 10.1. Equip V_0 with $\langle F, G\rangle_0 = 2\pi F^{\mathsf{T}}G$. Let $\mathcal{E}_{t,w}$ be the span of the first m positive surface eigenspaces. There is a resulting continuous isometry $U_{t,w} : V_0 \to \mathcal{E}_{t,w}$, obtained from collar-harmonic test functions, low-energy spectral projection, and polar decomposition. Define the pulled-back operator $Q_{t,w}$ by*

$$\langle Q_{t,w}F, G\rangle_0 = \int_{X_\Gamma(t,w)} \langle \nabla U_{t,w}F, \nabla U_{t,w}G\rangle.$$

Then

$$\sup_{w\in C}\left\| \frac{2\pi^2}{t}Q_{t,w} - L_\Gamma(w)|_{V_0}\right\| \longrightarrow 0 \qquad (t \downarrow 0). \tag{46}$$

For isolated clusters, the corresponding Kato-transported block forms also converge uniformly.

Proof. Let $J_{t,w}F$ equal the constant F_v on the thick core of the component indexed by v, interpolate harmonically across each collar, and subtract its mean. This is the standard test embedding in the graph-degeneration argument of Colbois–Colin de Verdière [10, Lemma V.1]. Uniformly for $w \in C$, collar capacity and the area of a pair of pants give

$$\langle J_{t,w}F, J_{t,w}G\rangle_{L^2} = 2\pi F^{\mathsf{T}}G + o(1)\|F\|\|G\|,$$
$$\int \langle \nabla J_{t,w}F, \nabla J_{t,w}G\rangle = \frac{t}{\pi}F^{\mathsf{T}}L_\Gamma(w)G + o(t)\|F\|\|G\|.$$

Indeed, a collar of core length tw_e has capacity $tw_e/\pi + o(t)$, and all thick data range in a compact set.

Let $\widehat{J}_{t,w} = J_{t,w}(J_{t,w}^* J_{t,w})^{-1/2}$ be the polar normalization of the test embedding, and let $R_{t,w}$ be the operator on V_0 obtained by pulling the Dirichlet form back through $\widehat{J}_{t,w}$. The two displayed estimates give

$$\sup_{w\in C} \left\| R_{t,w} - \frac{t}{2\pi^2} L_\Gamma(w)|_{V_0} \right\| = o(t). \tag{47}$$

The test space has dimension m and Rayleigh quotient $O(t)$. Hence the min–max principle gives m eigenvalues bounded by $O(t)$, while Otal–Rosas gives $\lambda_{m+1} > 1/4$. If $P_{t,w}$ is the orthogonal projection onto $\mathcal{E}_{t,w}$, the resulting uniform gap estimate makes $B_{t,w} = P_{t,w}\widehat{J}_{t,w}$ invertible for small t; explicitly,

$$\|(I - P_{t,w})\widehat{J}_{t,w}F\|_{L^2}^2 \le 4 \int |\nabla \widehat{J}_{t,w}F|^2 = O(t)\|F\|_0^2.$$

Take the polar part

$$U_{t,w} = B_{t,w}(B_{t,w}^* B_{t,w})^{-1/2},$$

where the adjoint uses $\langle\cdot,\cdot\rangle_0$ on V_0. The stronger small-eigenvalue form comparison of [10, Lemma V.1], applied uniformly on C, gives

$$\sup_{w\in C} \|Q_{t,w} - R_{t,w}\| = o(t). \tag{48}$$

This is the step that controls the energy of the discarded high-mode component; the preceding gap estimate alone only proves invertibility. Combining (47) and (48) proves (46). The contour formula for Riesz projections then gives uniform convergence of every isolated Kato block [17]. □

Let $A = L_\Gamma(w^0)|_{V_0}$ have distinct eigenvalues $\alpha_1 < \cdots < \alpha_r$ and eigenspaces E_i. Kato transport identifies the nearby cluster spaces with E_i and gives a block map

$$\mathcal{Q}_\Gamma(w) \in \bigoplus_{i=1}^{r} \mathrm{Sym}(E_i).$$

Following the essential-family terminology, we say that the family has the *weak Arnold hypothesis* (WAH) at w^0 if the target $\mathcal{Q}_\Gamma(w^0)$ lies in the interior of the image of every sufficiently small continuous perturbation of this block map on some compact neighborhood; compare the stable multiplicity framework of [12]. Surjectivity of

$$D\mathcal{Q}_\Gamma(w^0) : \mathbb{R}^{E_{\mathrm{eff}}(\Gamma)} \longrightarrow \bigoplus_i \mathrm{Sym}(E_i) \tag{49}$$

is the strong Arnold hypothesis (SAH) and implies WAH. Indeed, choose a linear complement S to the kernel of the derivative in (49). On the affine slice $w^0 + S$ the derivative is an isomorphism. A sufficiently small closed ball in this slice lies in the positive cone and has nonzero local Brouwer degree; every sufficiently small C^0 perturbation has the same boundary degree and therefore still hits the target.

Theorem 12.2 (Essential maximal-cusp transfer)**.** *If a trace-one graph direction p has a strict WAH realization on a genus-g all-pants dual graph, then there is $s_p > 0$ such that every sp, $0 < s < s_p$, is exactly the first m positive-eigenvalue list, with all multiplicities preserved.*

Proof. Choose the compact neighborhood in the WAH definition inside the positive weight cone and shrink it if necessary. By Lemma 12.1, the rescaled surface block map is uniformly C^0-close there to the graph block map for every sufficiently small t. WAH therefore supplies a positive nearby weighting at which the surface block form hits the target blocks exactly. The eigenvalues of the rescaled low operator $(2\pi^2/t)Q_{t,w}$, including all multiplicities, are therefore p, so the actual

surface eigenvalues are $(t/(2\pi^2))p$. Putting $t = 2\pi^2 s$ gives every sufficiently small multiple sp; the pointwise Otal–Rosas theorem places the next eigenvalue above $1/4$. □

The SSPWL is sufficient for the hypothesis above. Indeed, [8, Proposition 2.9] identifies SSPWL with transversality of the graph-pattern space to the orthogonal isospectral orbit. The normal quotient to that orbit is $\bigoplus_i \mathrm{Sym}(E_i)$, so the projected derivative is exactly (49). Thus SSPWL implies SAH and WAH. This does not say that the lifted surface itself has SAH.

If the positive multiplicities are $r_1, \dots, r_k$, a useful necessary condition follows by counting dimensions:

$$\sum_i \frac{r_i(r_i+1)}{2} = m + \sum_i \binom{r_i}{2} \le 3g-3, \qquad \text{hence} \qquad \boxed{\sum_i \binom{r_i}{2} \le g.} \tag{50}$$

Loops and parallel cuffs can only reduce the number of independent effective forms, so (50) is not sufficient.

13. The sharp cusp variance and forbidden cone

Let $b_m = (1/m, \dots, 1/m)$ and $V(p) = \|p - b_m\|_2^2$. For a trace-one effective graph write q for the number of positive distinct nonloop pairs, $d_v = \sum_{e \ni v} w_e$, and $W = \sum_e w_e = 1/2$. Then

$$\operatorname{tr} L^2 = \sum_v d_v^2 + 2\sum_e w_e^2 \ge \frac{1}{N} + \frac{1}{2q} \ge \frac{4}{3N}, \tag{51}$$

because $q \le 3N/2$. This includes disconnected positive supports and hence the multiscale boundary.

More precisely, (51) has the following exact quadratic SOS form:

$$\boxed{\operatorname{tr} L^2 - \frac{4}{3N} = \sum_v \left(d_v - \frac{1}{N}\right)^2 + 2\sum_{e=1}^{q} \left(w_e - \frac{1}{3N}\right)^2 + \frac{3N-2q}{9N^2}.} \tag{52}$$

The last slack is essential unless $q = 3N/2$. Equality occurs precisely for a loopless simple cubic effective support with all weights $1/(3N)$. Weighted-Laplacian minimum-variance inequalities and the equal-weight regular-graph minimizer already appear in [15, Section 4]. The point here is the optimization over every stable genus-g type, including zero-weight faces, and its sharp transfer to the surface limit-inf.

Theorem 13.1 (Sharp maximal-cusp variance)**.** *For $g \ge 3$, with $N = 2g-2$ and $m = N-1$,*

$$\boxed{\lim_{\varepsilon \downarrow 0} \inf_{\lambda_m(X) \le \varepsilon} V(p(X)) = c_g := \frac{N-4}{3N(N-1)} = \frac{g-3}{3(g-1)(2g-3)}.} \tag{53}$$

For every $g \ge 4$ and every $r < \sqrt{c_g}$, there is $\varepsilon_r > 0$ such that

$$\lambda_m(X) \le \varepsilon_r \quad \Longrightarrow \quad \|p(X) - b_m\|_2 \ge r.$$

The radius $\sqrt{c_g}$ is sharp at the tangent-link level.

Proof. For $p \in \mathcal{T}_g$, trace normalization gives $\sum_j p_j^2 = \operatorname{tr} L^2$. Hence (51) yields

$$V(p) = \sum_j p_j^2 - \frac{1}{N-1} \ge \frac{4}{3N} - \frac{1}{N-1} = c_g.$$

The bound is attained by the equal weighting of K_4 when $N = 4$ and, for every even $N \ge 6$, by an equal-weight connected simple cubic graph, for example the prism $C_{N/2} \,\square\, K_2$. Such a graph is an all-pants dual graph of genus $N/2 + 1 = g$, so Theorem 9.2 makes the bound sharp on the link.

To pass from the link to (53), take an arbitrary sequence with $\lambda_m \to 0$ and then a convergent subsequence of its normalized directions; every limit lies in $\mathcal{T}_g$. Conversely, a surface sequence approaches every link point by definition. Compactness of the link gives equality of the limit-inf and $\min_{\mathcal{T}_g} V$. The finite-scale exclusion for every smaller radius follows immediately. □

For $g = 2$ the link consists of one normalized coordinate and $V = 0$; the formula for c_g is not used. For $N > 4$, an equal-weight simple cubic graph with simple Laplacian spectrum supplies a genuine skeleton point. Indeed, its direction minimizes V globally on $\mathcal{T}_g$. A regular preimage would make that direction interior, but moving slightly toward b_m lowers V below c_g. Thus no stable graph realization of that direction is normalized regular. The skeleton is therefore not merely a defect of the Sard argument.

Corollary 13.2 (A cusp condition-number bracket)**.** *For $g \geq 3$, put $N = 2g-2$, $m = N-1$, and*

$$\kappa_g^* = \inf_{p \in \mathcal{U}_g} \frac{p_m}{p_1}, \qquad q_g = \frac{N-4}{3N}.$$

Then

$$\left(\sqrt{1+q_g} + \sqrt{q_g}\right)^2 \leq \kappa_g^* < 18 + 12\sqrt{2}. \tag{54}$$

Equivalently, the lower bound is

$$\frac{5N - 8 + 4\sqrt{(N-1)(N-4)}}{3N}.$$

It tends to 3 as $g \to \infty$. The bounds do not determine the optimum.

Proof. Let $p \in \mathcal{T}_g$ be positive and put $R = p_m/p_1$. For numbers $x \in [a,b]$ with mean μ, averaging $(x-a)(b-x) \geq 0$ and optimizing the result over $\mu/a \in [1, b/a]$ gives

$$\frac{\mathrm{mean}(x^2)}{\mu^2} \leq \frac{(R+1)^2}{4R}.$$

Apply this to the m coordinates of p. Since $\sum_j p_j = 1$,

$$1 + mV(p) = m\sum_j p_j^2 \leq \frac{(R+1)^2}{4R},$$

and hence

$$mV(p) \leq \frac{(R-1)^2}{4R}.$$

Theorem 13.1 gives $mV(p) \geq mc_g = q_g$. Inverting the last inequality on the branch $R \geq 1$ gives the left side of (54). The regular Ramanujan direction constructed in Theorem 11.1 gives the strict upper bound. The alternative formula and the limit are direct algebra. This corollary is a range reformulation of the sharp variance bound, not a separate optimality theorem. □

14. The classical genus-three prescription as endpoint calibration

The vanishing of c_3 reflects exact flexibility rather than only failure of the obstruction.

Theorem 14.1 (Genus-three prescription)**.** *For every $0 < a \leq b \leq c$ there is $s_0 > 0$ such that, for every $0 < s < s_0$, a closed genus-3 hyperbolic surface satisfies*

$$(\lambda_1, \lambda_2, \lambda_3) = s(a,b,c), \qquad \lambda_4 > \frac{1}{4}.$$

Repeated entries are preserved exactly.

Proof. On K_4 put

$$z = \frac{a}{4}, \qquad y = \frac{b-z}{3}, \qquad x = \frac{c-y-z}{2},$$

and assign

$$w_{12} = x, \qquad w_{13} = w_{23} = y, \qquad w_{14} = w_{24} = w_{34} = z.$$

The inequalities $0 < a \le b \le c$ give $x, y, z > 0$. The mutually orthogonal vectors of types $(1,1,1,-3)$, $(1,1,-2,0)$, and $(1,-1,0,0)$ show that the positive spectrum is

$$(4z, 3y+z, 2x+y+z) = (a,b,c).$$

Moreover, the linear map

$$\mathbb{R}^{E(K_4)} \longrightarrow \mathrm{Sym}(\mathbf{1}^\perp), \qquad w \longmapsto L_{K_4}(w)|_{\mathbf{1}^\perp}, \tag{55}$$

is an isomorphism between six-dimensional spaces. On functions modulo constants use coordinates $r_i = f_i - f_4$, $1 \le i \le 3$. The three edge forms for $i4$ are r_i^2, while those for ij are $(r_i - r_j)^2$; together they span every quadratic form in (r_1, r_2, r_3). Thus this strict family has SAH, including at collisions.

Put $S = a+b+c$ and $p = (a,b,c)/S$, and divide the displayed edge weights by S. This gives the trace-one direction required by Theorem 12.2. That theorem produces the exact surface form for every sufficiently small multiple of p; reparameterizing the scale by S gives every sufficiently small $s(a,b,c)$. Form equality, rather than differentiability of ordered eigenvalues, preserves the multiplicities. Finally use Otal–Rosas for λ_4. □

The classical antecedent is property A_4 of Colin de Verdière [11, Theorem 1.3 and §2a]: in genus three it prescribes the zero mode and three positive eigenvalues, with multiplicities, using metrics that may be chosen of constant curvature. Because its curvature/area normalization is not fixed, it is not by itself the curvature--1, all-sufficiently-small-scale statement above. Our proof is a direct K_4 normalization of the classical essential-family mechanism. We claim no novelty for that mechanism; the contribution used elsewhere in the paper is the fixed-cusp formulation together with the surrounding tangent-link and sharp genus-3/4 transition.

15. Semialgebraic complexity and SOS certificates

For rational ordered $p = (p_1, \dots, p_m)$ with $p_j \ge 0$ and $\sum_j p_j = 1$, define

$$\mathsf{MCTL} = \{(g,p) : p \in \mathcal{T}_g\}.$$

Theorem 15.1 (Complexity upper bounds)**.** *One has*

$$\mathsf{MCTL} \in \exists\mathbb{R} \subseteq \mathrm{PSPACE}.$$

For every fixed g, rational membership in $\mathcal{T}_g$ is decidable in deterministic time polynomial in the total bit length of p; the polynomial and its constants may depend on g.

Proof. It is not enough, when g is part of the input, to enumerate all stable graphs. Instead give each of the N labeled vertices three half-edges and use Boolean real variables $x_{\{a,b\}}$, indexed by unordered pairs of distinct half-edges, to encode a perfect matching:

$$x_{\{a,b\}}^2 - x_{\{a,b\}} = 0, \qquad \sum_{b \ne a} x_{\{a,b\}} = 1.$$

A standard single-commodity flow of polynomial size enforces connectedness of the matched pseudograph. Give a matched nonloop pair a variable $0 \le w_{\{a,b\}} \le x_{\{a,b\}}/2$, set loop energy to zero, and impose $2\sum w_{\{a,b\}} = 1$ over nonloop pairs. Parallel terms simply add in the resulting Laplacian L. A connected cubic pseudograph on N vertices has first Betti number $3N/2 - N + 1 = g$. A stable vertex of type $(h, \deg) = (1,1)$ is represented by a genus-zero cubic vertex whose two

additional half-edges are paired into a self-loop; this changes neither the effective graph energy nor its spectral image. Thus the cubic encoding covers every type in $\mathfrak{G}_g$, and zero conductances recover all of their weight faces.

Introduce $U \in \mathbb{R}^{N\times m}$ and impose

$$U^{\mathsf{T}}U = I_m, \qquad U^{\mathsf{T}}\mathbf{1} = 0, \qquad L = U\,\mathrm{diag}(p_1,\dots,p_m)U^{\mathsf{T}}. \tag{56}$$

These are polynomially many quadratic equations and inequalities, giving a polynomial-size existential-real formula. Canny's theorem yields the PSPACE upper bound [7]. When g is fixed, the real dimension, degrees, and number of constraints are fixed, so Renegar's bounds give polynomial bit complexity [24]. No hardness or uniform fixed-parameter claim is made. □

Regular membership also has a polynomial existential witness. If $J_\Gamma(w)$ is the m-row unnormalized coefficient Jacobian, use a right inverse variable

$$J_\Gamma(w)R = I_m \tag{57}$$

instead of expanding exponentially many minors.

There is a complementary graph-by-graph certificate for nonmembership. For a fixed effective type with q edges put

$$K_\Gamma = \{w \geq 0 : 2\sum_e w_e = 1\}$$

and, for trace-one p, define

$$F_{\Gamma,p}(w) = \sum_{k=2}^{N}\left(\operatorname{tr} L_\Gamma(w)^k - \sum_{j=1}^{m} p_j^k\right)^2. \tag{58}$$

The trace fixes the first power sum, and Newton identities show

$$p \in \psi_\Gamma(K_\Gamma) \quad\Longleftrightarrow\quad \min_{K_\Gamma} F_{\Gamma,p} = 0. \tag{59}$$

If p is not in this graph image, compactness makes the minimum $\delta_{\Gamma,p} > 0$. Add the redundant ball inequality $b(w) = 1 - \|w\|^2 \geq 0$ and let $h(w) = 1 - 2\sum_e w_e$. The resulting quadratic module is Archimedean. For $0 < \epsilon < \delta_{\Gamma,p}$, Putinar's Positivstellensatz gives, at some finite degree,

$$F_{\Gamma,p} - \epsilon = \sigma_0 + \sigma_b b + \sum_e \sigma_e w_e + \tau h, \tag{60}$$

where the σ's are sums of squares and τ is arbitrary [23]. Equivalently, some finite Lasserre relaxation separates the optimum from zero [19]. Since the stable types are finite for fixed g, a nonmember of $\mathcal{T}_g$ has one such certificate for every type.

This conclusion has three important limits. There is no uniform bound on the SOS degree or bit size; the coefficients in (60) may be real; and membership or boundary nonnegativity is not certified by the same strict-positivity argument. In particular, this does not place the complement in NP or make the Lasserre hierarchy a two-sided decision procedure.

16. Residual directions and the global problem

The preceding theorems leave a lower-dimensional but genuine residual set. Two cubic graphs make the boundary of the argument explicit.

16.1. **The genus-four collision barrier.** The unit-weight Laplacian of $K_{3,3}$ has spectrum

$$(0,3,3,3,3,6), \qquad p_{3,3} = \left(\frac{1}{6},\frac{1}{6},\frac{1}{6},\frac{1}{6},\frac{1}{3}\right), \qquad V(p_{3,3}) = c_4 = \frac{1}{45}.$$

Its positive multiplicities are $4+1$, so the block-form target dimension is $10+1=11$, whereas a genus-4 pants family has at most $3g-3=9$ independent cuff parameters. Hence no leading graph realization of this pattern can satisfy WAH or SAH. This is only a first-order barrier: it proves neither realization nor nonrealization of the exact surface ray.

16.2. **The genus-seven Frucht fold.** Let F be the Frucht graph, with 12 vertices and 18 edges. Its adjacency characteristic polynomial factors as

$$(x-3)(x-2)x(x+1)(x+2)(x^3+x^2-2x-1)(x^4+x^3-6x^2-5x+4). \tag{61}$$

The two nonlinear factors have discriminants 49 and 22545, resultant 64, and no roots among $3,2,0,-1,-2$. Thus the equal-weight Laplacian has simple positive spectrum. Its normalized direction p_F satisfies $V(p_F) = c_7 = 2/99$ and is a global minimizer of V on $\mathcal{T}_7$, hence a critical value of the normalized graph map.

Proposition 16.1 (Certified definite Frucht fold)**.** *At unit weights, the unnormalized coefficient Jacobian of F has rank* 10 *and contains a minor of determinant* 73728*. On the fixed-total-weight simplex, the restriction of the coefficient Jacobian has rank* 9 *and contains a minor of determinant* 36864*. Consequently the normalized spectral derivative has rank* 9*, and the normalized spectral map has a corank-one definite fold at p_F.*

Proof. The accompanying exact-arithmetic certificate computes the characteristic coefficients through the multiaffine weighted-forest formula and verifies the two integer minors. The variance boundary supplies the matching upper ranks 10 and 9, so the computed ranks are exact. On the fixed-weight-sum slice let F_0 be the normalized spectral map and $h = V - c_7$. Then $dh(p_F) = 2(p_F - b_m)$ spans the one-dimensional cokernel. The cubic equality case of (52) gives, with $W = \sum_e w_e$, $\bar{d} = 2W/12$, and $\bar{w} = W/18$,

$$V(F_0(w)) - c_7 = \frac{\sum_v (d_v - \bar{d})^2 + 2\sum_e (w_e - \bar{w})^2}{4W^2},$$

whose Hessian is positive definite on the eight-dimensional kernel of DF_0. The Morse lemma with parameters gives local coordinates

$$(x,y) \longmapsto (x, \|y\|^2), \qquad x \in \mathbb{R}^9, \quad y \in \mathbb{R}^8.$$

The underlying cubic graph appears in the catalogue [6]; the displayed factorization and both minors are independently verified by the accompanying exact-arithmetic certificate. The definiteness of the fold is supplied analytically by the displayed SOS identity, not by the rank certificate alone. □

This proposition is a graph theorem, not an exact surface-ray theorem. Burger's C^0 comparison does not determine which side of a fold the surface perturbation occupies. If a fixed plumbing family admits a refined uniform expansion

$$F_t = F_0 + a(t)R + o_{C^2}(a(t)), \qquad a(t) \downarrow 0,$$

Lyapunov–Schmidt reduction leaves a scalar normal equation

$$\|y\|^2 + a(t)\beta_F + o(\|y\|^2 + a(t)) = 0.$$

The sign of the still-uncomputed coefficient β_F decides local existence in that family. It does not by itself give a global exclusion over all twists and graph-limit representations.

16.3. **Open ends.** The exact link, generic simple directions, and WAH collision directions do not solve the finite-scale problem of [14, Question 1]. Among the remaining concrete problems are:

(1) decide the $K_{3,3}$ and Frucht surface rays, the latter by computing the signed normal plumbing correction;
(2) build a compactified spectral cobordism controlling nonseparating Deligne–Mumford escape, eigenvalue collisions, and crossings at $1/4$;
(3) determine the optimal universal aperture δ_n^* and the true asymptotic order of the path volume $P(P_n)$.
(4) determine the exact regular cusp condition number κ_g^* between the bounds in Corollary 13.2, and describe its minimizing graph types.

Thus the maximal-cusp tangent link is completely characterized, and its exact surface germ is settled off a compact semialgebraic null skeleton and on all essential collision strata. Inside the regular germ there is, for every genus, a genus-dependent full-dimensional wedge whose internal condition number is bounded by the genus-independent constant $18+12\sqrt{2}$ and whose every target has continuum many realizations; common lacunary targets also persist across superexponentially many all-pants thin regions. The residual skeleton, optimal condition number, and global inverse problem remain open.

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